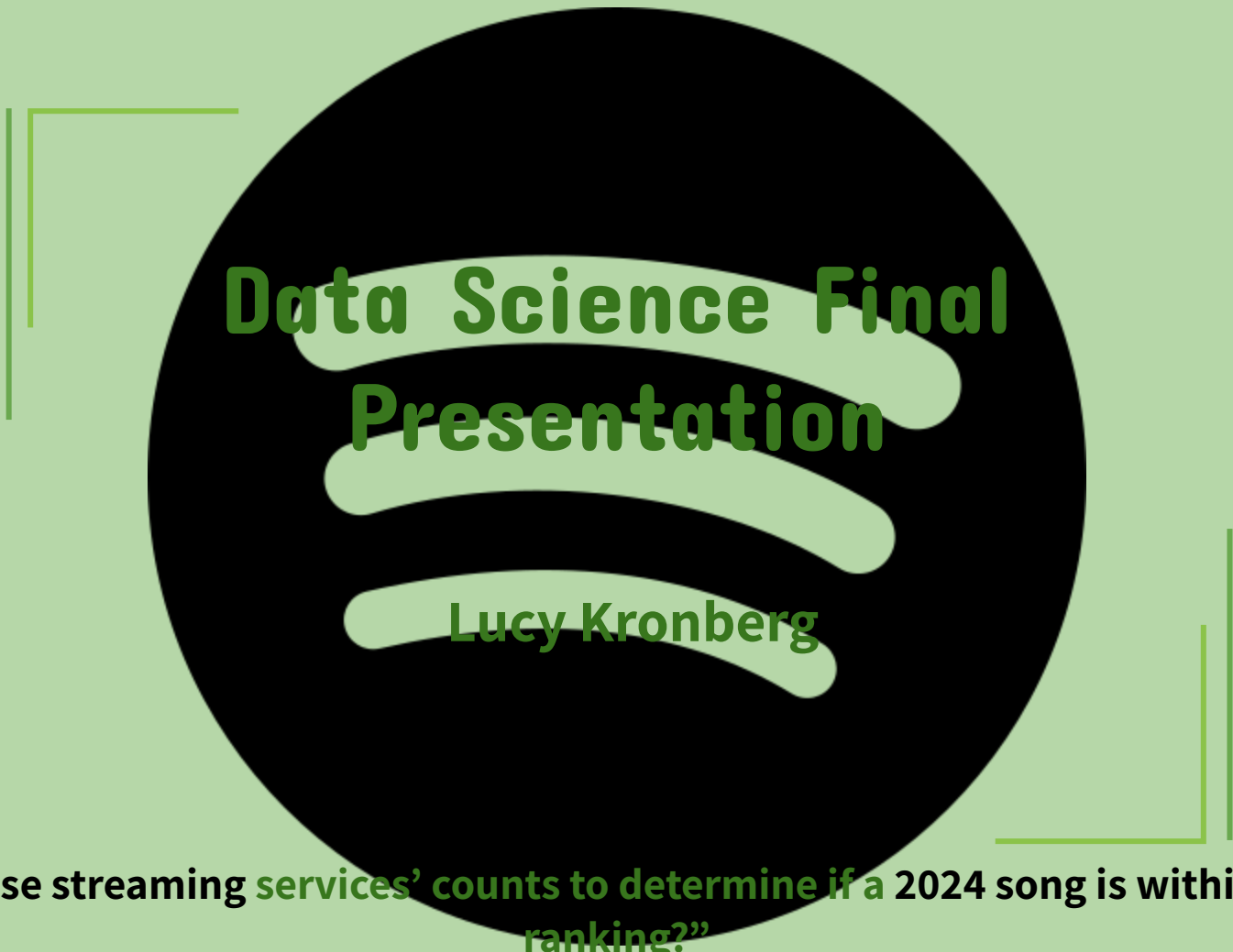


A large black circle is centered on the slide. To its left, a green line starts from the top, goes down, and then turns right. To its right, a green line starts from the middle, goes down, and then turns left. The text is overlaid on the circle.

Data Science Final Presentation

Lucy Kronberg

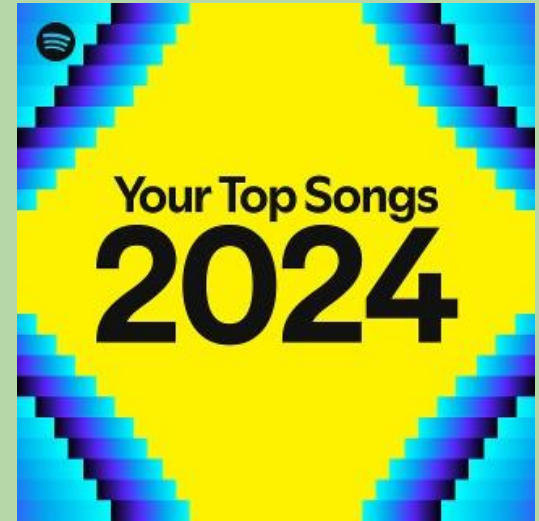
“Can you use streaming **services’ counts to determine if a 2024 song is within a top 500 ranking?**”

Pt. 1 - Dataset

Dataset: Most Streamed Spotify Songs 2024

Dataset from: [kaggle.com](https://www.kaggle.com)

Includes: Data collected of popular 2024 songs by Nidula Elgiriye withana from streaming services such as Spotify, YouTube, Deezer, Apple Music, Shazam, TikTok, and more!



Dataset Overview:



Track	Album Name	Artist	Release Date	All Time Rank	Track Score	Spotify Streams	Spotify Playlist Count	Spotify Playlist Reach	Pe
MILLION DOLLAR BABY	Million Dollar Baby - Single	Tommy Richman	4/26/2024	1	725.4	390,470,936	30,716	196,631,588	
Not Like Us	Not Like Us	Kendrick Lamar	5/4/2024	2	545.9	323,703,884	28,113	174,597,137	
i like the way you kiss me	I like the way you kiss me	Artemas	3/19/2024	3	538.4	601,309,283	54,331	211,607,669	
Flowers	Flowers - Single	Miley Cyrus	1/12/2023	4	444.9	2,031,280,633	269,802	136,569,078	
Houdini	Houdini	Eminem	5/31/2024	5	423.3	107,034,922	7,223	151,469,874	
Lovin On Me	Lovin On Me	Jack Harlow	11/10/2023	6	410.1	670,665,438	105,892	175,421,034	

Apple Music Playlist Count	AirPlay Spins	SiriusXM Spins	Deezer Playlist Count	Deezer Playlist Reach	Amazon Playlist Count
210	40,975	684	62	17,598,718	114
188	40,778	3	67	10,422,430	111
190	74,333	536	136	36,321,847	172
394	1,474,799	2,182	264	24,684,248	210
182	12,185	1	82	17,660,624	105
138	522,042	4,654	86	17,167,254	152

Song data:

Song Name:

Pt. 2 - Variables + Biases

Numerical Values: Apple Music Playlist Count and Deezer Playlist Count

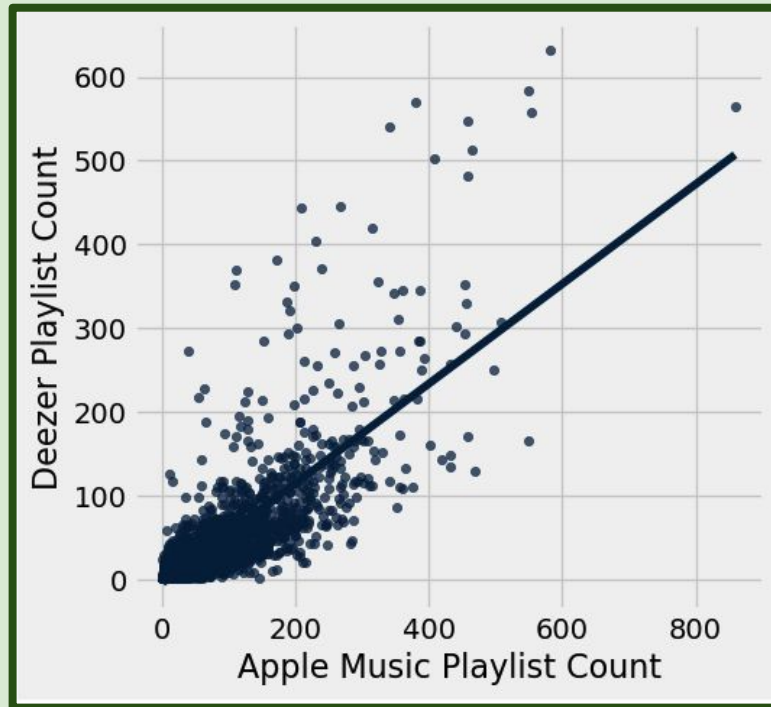
95% Confidence Interval: [0.7632038297002663, 0.8167997214704301]

99% Prediction Interval: [57.24590718127515, 64.66716850797047]

X-Input Prediction Value: 200

Sources of bias: my dataset is now 2 years outdated and doesn't include any new songs past 2024. Also, the creation of this dataset had little to no description on where or how the data was collected.

Apple Music Playlist Count v. Deezer Playlist Count Scatterplot



Correlation: 78.9%

Pt. 3 - Classifiers

Classifiers #1: 'Spotify Streams', 'Spotify Playlist Count', 'Spotify Playlist Reach', & 'Spotify Popularity'

Applying a classification function to my test dataset (roughly 700 rows of songs instead of over 2000) gave me an accuracy of13%. 🤖

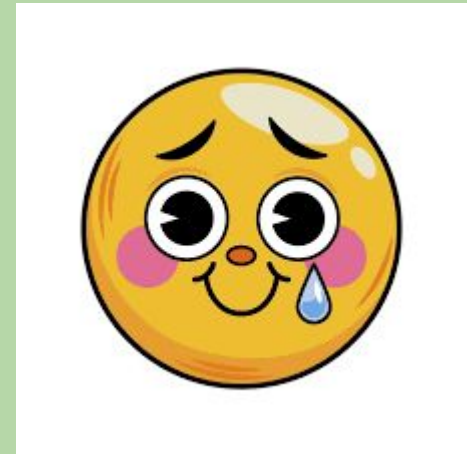
So I tried again!

Classifiers #2: 'Spotify Playlist Count', 'Amazon Playlist Count', 'Deezer Playlist Count', & 'Apple Music Playlist Count'

Using the same method I used for my first classifier, only now with new features, I was able to raise my song classification accuracy up to ...15%! Hold up.

HOW TO BUILD A CLASSIFIER:

1. Convert row to array (row_to_array)
2. Compare the nearest neighbor distances from LRL
3. Vote on the class! (must be odd)



15??????



Classifiers... take 2!

Changes: used original features,
sorted nearest neighbors in
descending = False order

$$X_{su} = (X - X_{avg}) / X_{sd}$$

Final accuracy: ~88%!!!!

Old dataset

Track	Album Name	Artist	Date	Rank	Score	Spotify Streams	Spotify Playlist Count	Spotify Playlist Reach	Popularity
MILLION DOLLAR BABY	Million Dollar Baby - Single				25.4	390,470,936	30,716	196,631,588	
Not Like Us	Not Like Us	Kendrick Lamar	5/4/2024	2	545.9	323,703,884	28,113	174,597,137	
i like the way you kiss me	I like the way you kiss me	Artemas	3/19/2024	3	538.4	601,309,283	54,331	211,607,669	
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Spotify Streams	Spotify Playlist Count	Spotify Playlist Reach	Spotify Popularity	Apple Music Playlist Count	Deezer Playlist Count	Deezer Playlist Reach	Amazon Playlist Count	Shazam Counts
2.20687	1.72431	2.16696	0.743548	433	149	1,452,991	97	14056461
1.7042	1.41457	0.421826	0.135529	291	98	937,524	94	19858741
1.91311	2.34932	1.06466	0.309249	153	111	467,038	77	14002280
1.47941	1.78221	-0.0174421	0.135529	67	34	183,823	33	7997160
4.05763	2.23164	2.23164	0.135529	2	3	3,006	89	15254141

Standardized data

Pt. 4 - Conclusion

“Can you use streaming services’ counts to determine if a 2024 song is within a top 500 rank?”

Answer: Yes! (well, mostly)

What did I learn?

- I learned that you can somewhat accurately predict whether or not a 2024 song was in the top 500 ranking.
- Based off of my classifiers I am guessing that Spotify (the likely creator of the track ranking system) uses these sort of classifiers to calculate how popular a song is currently (or in 2024 in this case)
- This dataset has ton of potential!